

Test 35: Spectral form factor K(t)

Verdict: FAIL | Generated: 2026-09-02 22:36:30 | Suite: AB-Cloud v19 (Julia 1.12.0)

What this test verifies

Statistical: $\text{raw } (1/n) | \langle e^{it} \rangle |^2$ vs finite-box GUE reference $K_{\text{box}}(t) = t(1 - \text{sinc}(2t))$; RMS over $t \in [0, 0.3]$.

METHOD: Statistical: $\text{raw } (1/n) | \langle e^{it} \rangle |^2$ vs finite-box GUE reference $K_{\text{box}}(t) = t(1 - \text{sinc}(2t))$; RMS over $t \in [0, 0.3]$.

Test 35 HARDCORE pass 2: $n=5$ per-realization $K(t)$, 33-point t -grid, ensemble mean verdict.

Test 35 HARDCORE: ensemble RMS=0.6143, corr_box=0.6379, corr_?=0.8123, RMS scatter ?=0.1856.

Test 35b [HARDCORE pass 2]: 3 sub-checks, 2 failed ? WARN

Result

- 'plots/plot_01' (.png 600dpi / .svg / .pdf)

Plots

- 'plots/plot_02' (.png 600dpi / .svg / .pdf)

- 'plots/animation.gif' ? animated reveal of every plot of this test

Reproduce with: `julia ab_cloud_v19.jl --test 35 --no-two-pass`

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

Independent verification guide

[22:09:09.750] [+30065.032s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=1.00, open, 2 vortices, model=:monumental

[22:14:37.106] [+30392.388s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=1.00, open, 2 vortices, model=:monumental

[22:19:44.711] [+30699.993s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=1.00, open, 2 vortices, model=:monumental

[22:24:48.414] [+31003.696s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=1.00, open, 2 vortices, model=:monumental

[22:30:51.804] [+31367.086s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=1.00, open, 2 vortices, model=:monumental

??

TEST 35 (pass 2, HARDCORE): spectral form factor $K(t)$? ensemble pass

CONSOLE CAPTURE

n=5 per-realization $K(t)$, 33-point t -grid, ensemble verdict

??

? RUNTIME: $n=5$ realizations at 96x96 ? expect ? 12?17 min

real 1/5: RMS($K ? K_{\text{box}}$, $t \in [0, 0.3]$) = 0.9199

real 2/5: RMS($K ? K_{\text{box}}$, $t \in [0, 0.3]$) = 0.8578

real 3/5: RMS($K ? K_{\text{box}}$, $t \in [0, 0.3]$) = 0.9671

real 4/5: RMS($K ? K_{\text{box}}$, $t \in [0, 0.3]$) = 0.7540

real 5/5: RMS($K ? K_{\text{box}}$, $t \in [0, 0.3]$) = 0.4999

ENSEMBLE K(t): RMS(K_AB ? K_box, t ? 0.3) = 0.6143 (target < 0.30)

Corr(K_AB, K_box) = 0.6379 (target > 0.50) | Corr(K_AB, K_?) = 0.8123

Per-realization RMS: mean 0.7998, scatter ? = 0.1856 > ? 0.15

??

RMS < 0.30 (ensemble) WARN ? ensemble-mean K(t), 33-point grid: 0.6143

Corr > 0.50 (ensemble) PASS ? Corr(K, K_box) = 0.6379

RMS scatter ? 0.15 WARN ? ?=0.1856 across 5 realizations

Test 35b [HARDCORE pass 2]: 3 sub-checks, 2 failed ? WARN